



CampusSafe Twin

A Campus Safety Digital Twin

Business Case Presentation

PRECIOUS MASETLA

Three Gaps in Campus Safety Today



Late or missing reports

Students lack a quick, accessible way to report incidents as they happen.



No consolidated visibility

Security staff have no real-time, spatial view of where incidents occur.



No route-level guidance

Students have no data-driven way to choose safer paths at night.

Result: incidents are under-reported, patrols are reactive, and students have no safety-informed way to navigate campus.

One Platform, Two Roles

A web-based digital twin built around an interactive campus map, with simulated data standing in wherever live university systems aren't available.



Student View

- Browse the interactive campus map
- Report incidents: category, location, description
- View safe-route suggestions



Security View (CPS)

- Live incident feed & map overlay
- Status management (new → resolved)
- Hotspot analytics & trend charts

OBJECTIVES

What the Prototype Sets Out to Do

-  Build an interactive digital campus map with incident markers
-  Enable role-based incident reporting for students
-  Propagate new reports to the dashboard in near real time
-  Deliver a dashboard with map + status management
-  Apply analytics to surface incident hotspots and trends
-  Recommend safer routes weighted by incident data

In Scope for This Prototype



Map & Reporting

- Interactive map of a defined campus area
- Incident reporting: theft, vandalism, suspicious activity, medical, other
- Optional photo attachment



Security Operations

- Live incident feed & map overlay
- Status workflow: new → acknowledged → in progress → resolved
- Filtering by category, date, status



Analytics & Routing

- Hotspot heatmap / clustering
- Trend charts by category & time
- Safe-route recommendation between two points



Platform Basics

- Student vs security role separation
- Basic authentication
- Simulated data generator for demos

Value Across Every Stakeholder Group



University / CPS

- Consolidated, map-based situational awareness
- Faster hotspot identification for patrol planning
- Higher-friction-free incident reporting



Students

- Quick, accessible reporting channel
- Awareness of recent incidents & risk areas
- Safer route suggestions at night



Project Team

- Hands-on digital twin & geospatial experience
- Strong, demonstrable capstone deliverable
- Full-stack + analytics + dashboard skills

The Biggest Risk — and How We Manage It



SCOPE CREEP BEYOND ONE SEMESTER

Third-year teams have limited time and a fixed skill set — the biggest threat to delivery isn't technical difficulty, it's trying to build too much.

Mitigation:

A strict, prioritised in-scope list — if time runs short, we cut features, not quality.



A Realistic, Achievable, Demonstrable Prototype

CampusSafe Twin applies digital twin thinking to a genuine campus safety challenge — an interactive map, incident reporting, a security dashboard, hotspot analytics, and safe-route guidance, scoped for one semester and grounded in simulated data where live integration isn't possible.

Thank you — questions welcome.